

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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TECHNICAL PROFILE

Embedded firmware engineer with experience in TI Embedded Linux platforms, a multi-controller firmware-update workflow, programmable power-conversion control, and CAN-connected battery systems. Experienced across C/C++, Linux services, board bring-up, bench debugging, and cross-platform Windows/Linux tooling.

TECHNICAL SKILLS

Firmware & OS: Embedded C/C++, Python, Bash; TI Embedded Linux, Yocto/Arago, Linux kernel/device trees, systemd, Docker, FreeRTOS

Processors & Interfaces: TI AM62, TI TMS320 DSP, PIC32, MSP430FR2355; CAN/CAN FD/J1939, I2C, SPI, UART/RS-232, ADC, PWM, GPIO, IPMI/IPMB

Development & Test: Git/GitLab, Code Composer Studio, JTAG, KiCad, SocketCAN/PCAN; oscilloscopes, logic/protocol analyzers, through-hole/SMT soldering

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Build and debug TI AM62-based Embedded Linux systems using Yocto/Arago images and toolchains; compile and integrate kernels and device trees, diagnose boot and service failures, and develop systemd services for external-watchdog monitoring and recovery.
- Integrate and debug APU/EPU embedded power subsystems spanning battery hardware, CAN-connected devices, and Linux services; trace faults across power and wiring, bus traffic, Linux configuration, service logs, and application code.
- Diagnose CAN/CAN FD, IPMI/IPMB, I2C, SPI, and UART behavior with logs, register reads, analyzers, scopes, schematics, datasheets, and controlled substitutions to isolate hardware, firmware, timing, and configuration faults.
- Develop C/C++, Python, and Bash software, use Docker in development workflows, and improve code and tooling portability across Windows and Linux.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected engineering across both internships

- Developed TMS320 DSP firmware controlling transformer-coupled bidirectional buck–boost power stages through 12 PWM A/B pairs (24 physical outputs): independently configured each A output's period/frequency, duty cycle, and relative phase; paired B matched or inverted A.
- Developed multiple connected stages of one PIC32/TMS320 product's firmware-update workflow: a user-facing C++ CLI accepted supplied HEX files, Python/TCP transport provided chunking and checksum validation, and the processors coordinated external SPI-flash/golden-image handling, UART DSP programming, verification, and boot control.
- In 2024, built a restricted SSH/chroot CLI with Python/Bash for TI Yocto/Arago embedded power hardware; reverse-engineered RS-232 transactions and checksums; supported CAN/FreeRTOS debugging, schematic review, soldering, and burn-in-test hardware.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Built an Excel/VBA tolerance checker for manufactured parts and organized a searchable 17,000+ record catalog connecting tooling, parts, vendors, operations, locations, and revisions.

SENIOR DESIGN

EV Active Sensor Adapter | MSP430FR2355

NC State University

- Defined end-to-end architecture, requirements, interface specifications, risks, and validation plans for sensing, BMS and power-management hardware, custom-PCB interfaces, embedded control, communications, and telemetry; coordinated decisions and documentation across the team.
- Evaluated BMS/support ICs against electrical, protocol, package, and sourcing constraints using datasheets and reference designs; translated selections into integration plans, prototyped interrupt-driven SPI, and worked with ADC, UART, timer, GPIO, and LCD interfaces.

TEACHING & TECHNICAL LEADERSHIP

ECE 306 Embedded Systems Instruction | NC State University

Spring 2025–Spring 2026

- Supported three semesters: two in an unofficial role at about 10 hr/week and one as the official TA at 20 hr/week; delivered supplemental lectures, test preparation, labs, code/PCB debugging, project guidance, and ECE-program support.
- Added supplemental lectures and intensive, question-driven support as the class average rose from its typical C+ to A-.

IEEE Open Project Space 2 | Lead Instructor

Fall 2025 & Spring 2026

- Ran two semesters of a project-based electronics course from idea through demonstration; taught firmware, PCB/BOM workflows, and through-hole/SMT assembly while coaching students to isolate and solve their own design problems.

IEEE Soldering Workshops | Coordinator, Lecturer & Lab Instructor

- Coordinated, lectured, and supervised standalone workshops using custom through-hole and surface-mount flashing-light PCB keepsakes; taught professional technique, safety, inspection, debugging, and rework.